CHAPTER 6

Differentiation, variations and optimisation

Chapter path

Objectives. Calculate a derivative on its domain; study variations and extrema; compare of curves; optimise; exploit symmetries.
Prerequisites. Derivative at a point, algebraic calculation, sign of a quadratic, intervals.
Yearly reference point. December 1·10: 8:00 a.m., consolidation on 15 and 17

Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. Derivative of x^2 at 3?

2. Domain of $\dfrac{1}{x-2}$?

3. Sign of $(x-1)(x+2)$?

4. Expand $(x^2+1)(x-2)$.

5. For $f(x)=x^2$, compare $f(-x)$ and $f(x)$.

6. If $f(1)=2$, $f'(1)=0$, tangent equation?

After the diagnostic check. Go over the flagged concepts with the teacher; redo a question similar, then explain the rule used.

Oral work and logic. Present a variation table, distinguishing the sign of f from the sign of its derivative. Explain why the endpoints of an optimisation problem must be examined.

Investigate before formalising

Can an increasing function have a horizontal tangent? Look for an example; can it have a strictly negative derivative at an interior point?

Organisation. Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min. Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.

6.1 Derivative function and standard functions

Definition and results

A function is differentiable on an interval when it is differentiable at each of its points; its derivative assigns to x the number $f'(x)$. The standard derivatives are: constant $\mapsto 0$; $x \mapsto 1$; $x^2 \mapsto 2x$; $x^3 \mapsto 3x^2$; $\frac{1}{x} \mapsto -\frac{1}{x^2}$ for $x \neq 0$; $\sqrt{x} \mapsto \frac{1}{2\sqrt{x}}$ for $x > 0$. For $n \in \mathbb{Z}$, $(x^n)' = nx^{n-1}$ on its domain of differentiability; for $n = 0$, the derivative is 0 directly.

Status and justification. Square: the rate is $2a + h$. Reciprocal: for $a \neq 0$, $h \neq 0$ and $a + h \neq 0$, $\frac{\left[\frac{1}{a+h} - \frac{1}{a}\right]}{h} = -\frac{1}{[a(a+h)]}$, which tends to $-\frac{1}{a^2}$. The other formulas are accepted here.

Worked example. For $f(x) = x^{-3}$, the domain is $\mathbb{R} \setminus \{0\}$ and $f'(x) = -3x^{-4} = -\frac{3}{x^4}$. For $g(x) = \sqrt{x}$, the function is defined at 0, but its usual derivative exists only for $x > 0$.

Watch out

The set on which a function is differentiable may be smaller than its domain of definition.

Immediate application. Differentiate x^5, x^{-2} and $\sqrt{x}$ stating the domains.

6.2 Sum, product, reciprocal and quotient

Definition and results

For u and v differentiable, $(u + v)' = u' + v'$, $(\lambda u)' = \lambda u'$, $(uv)' = u'v + uv'$. Where $v \neq 0$, $(\frac{1}{v})' = -\frac{v'}{v^2}$ and $(\frac{u}{v})' = \frac{u'v - uv'}{v^2}$.

Status and justification. For the product, add and subtract $u(a+h)v(a)$ in the numerator. The rate becomes $\frac{u(a+h)[v(a+h) - v(a)]}{h} + \frac{v(a)[u(a+h) - u(a)]}{h}$. Since a differentiable function is continuous (admitted), we obtain $u(a)v'(a) + v(a)u'(a)$. The other rules are accepted without proof.

Worked example. For $f(x) = \frac{x^2+1}{x-1}$, $x \neq 1$, $f'(x) = \frac{[2x(x-1) - (x^2+1)]}{(x-1)^2} = \frac{x^2 - 2x - 1}{(x-1)^2}$. Keeping the brackets avoids changing only one sign instead of the whole numerator.

Watch out

The derivative of a product is not the product of the derivatives; the derivative of a quotient is not their quotient.

Immediate application. Differentiate $(x^2+1)(3x-2)$ then $\dfrac{1}{x^2+1}$.

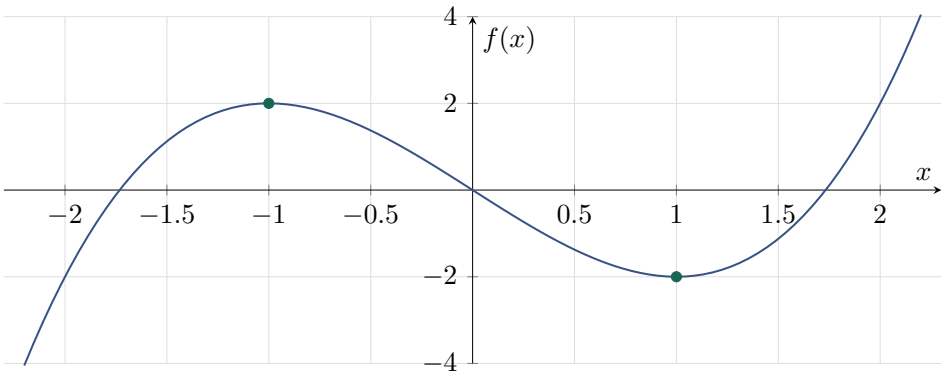
6.3 Sign of the derivative and extrema

Definition and results

On an interval I, if $f' \geq 0$, f is increasing; if $f' \leq 0$, it is decreasing; if $f' = 0$ everywhere, it is constant. A strictly positive derivative implies it is strictly increasing. If f has a local extremum at an interior point where it is differentiable, its derivative there is zero. The converse is false.

Status and justification. Variation theorems assumed. For an interior minimum at a, the rates on the right are positive or zero, those on the left negative or zero; if they converge to the same real number, it is 0. Same argument for a maximum.

Worked example. $f(x) = x^3 - 3x$: $f'(x) = 3(x-1)(x+1)$. f increases on $]-\infty; -1]$, decreases on $[-1; 1]$, increases on $[1; +\infty[$; local maximum $f(-1) = 2$, local minimum $f(1) = -2$. None of these extremum is global on $\mathbb{R}$.



$f(x) = x^3 - 3x$: a local maximum at -1 and a local minimum at 1.

Watch out

The theorem applies on an interval. A function may be decreasing on two intervals separate intervals without decreasing on their union.

Immediate application. Study $f(x) = x^2 - 4x + 1$ on $[0; 3]$ and give its extrema on this interval.

6.4 Parity, inequalities and comparison

Definition and results

On a domain symmetric about 0, f is even if $f(-x) = f(x)$ and odd if $f(-x) = -f(x)$. The graph of an even function is symmetric about the axis of y-axis; that of an odd function with respect to the origin. To compare f and g, we study the sign of $f - g$, possibly with its derivative.

Status and justification. The symmetries follow from the coordinates of the image points. For a inequality, a study of variations can give a minimum of the difference, then its sign; the derivative does not directly replace this difference.

Worked example. For $x > 0$, $h(x) = x + \frac{1}{x} - 2$ has derivative $\frac{x^2-1}{x^2}$. It decreases up to 1 then increases; $h(1) = 0$, therefore $x + \frac{1}{x} \geq 2$. Equality holds only at 1.

Watch out

Check the domain before parity. The sign of $f' - g'$ gives information about the variation of $f - g$, not directly from its sign.

Immediate application. Study the parity of $x^4 + x^2$ and $x^3 - x$; prove $x^2 + 1 \geq 2x$.

6.5 Optimisation and Newton's method

Definition and results

Optimising consists of defining the variable and its domain, expressing the quantity to be optimised, then compare all the candidate values, including the endpoints. Newton's method uses the tangent to approximate a solution of $f(x) = 0 : x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ when $f'(x_n) \neq 0$. It is used only in favourable cases.

Status and justification. Newton's formula comes from the intersection with the horizontal axis of $y = f(x_n) + f'(x_n)(x - x_n)$. Convergence is not guaranteed in general; it is observed and checked in the case chosen.

Worked example. For $f(x) = x^2 - 2$ with $x_0 = 1$, Newton's method gives $x_1 = \frac{3}{2}$, $x_2 = \frac{17}{12}$. The squares are 2.25 then about 2.00694: we approach $\sqrt{2}$. The program must check for a zero derivative and the number maximum of steps.

Watch out

A numerical value given by an algorithm is not proof of a minimum or of convergence.

Immediate application. A rectangle with perimeter 12 m has width x. Find the maximum area by differentiation.

Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher chooses one. Exercise 8 is revisited later.

Exercise 6.1 · A collection of derivatives

Consider the functions

$$f(x) = 3x^4 - 2x + 7, \quad g(x) = (x^2 + 2)(x - 1), \quad h(x) = \frac{2x + 1}{x + 3}.$$

1. State the domain of each of these functions.
2. Calculate their derivatives, naming the rule used each time. For g, first keep the two terms of the product rule; for h, write the numerator before simplifying it.
3. Check the result concerning g by expanding its expression before differentiating.

Exercise 6.2 · A complete table

The function f is defined on [0; 4] by $f(x) = x^3 - 6x^2 + 9x + 1$. We seek its minimum and maximum on this interval.

1. Calculate and factorise $f'(x)$, then study its sign on [0; 4].
2. Draw the variation table of f indicating the values at the bounds and at the points where the derivative is zero.
3. Give the minimum and maximum of f on the interval, as well as all the x-coordinates where they are reached.

Exercise 6.3 · A quotient

For every strictly positive real number x, set $f(x) = x + \frac{4}{x}$.

1. Study the variations of f on $]0; +\infty[$. Set out the factors that determine the sign of the derivative.
2. Deduce the smallest value of $x + \frac{4}{x}$, then write an inequality valid for every $x > 0$. State the equality case precisely.

Exercise 6.4 · Compare two curves

Functions $f(x) = x^3$ and $g(x) = x$ are defined on $\mathbb{R}$. Let $\mathcal{C}_f$ and $\mathcal{C}_g$ their curves in a same coordinate system.

1. Determine the intersection points of these two curves.

2. Study their relative position on $\mathbb{R}$ choosing a suitable method. Interpret your result by the intervals where $\mathcal{C}_f$ is above or below $\mathcal{C}_g$.

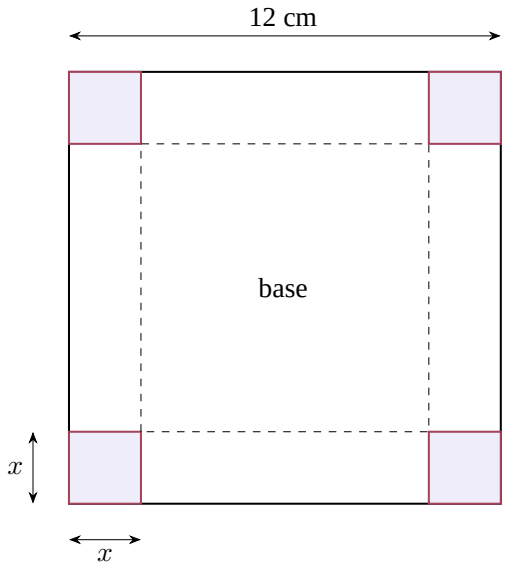
Exercise 6.5 · A package

An open box (no lid) is made from a square sheet of side 12 cm. We cut out in each of the four corners a square of side x cm, then the remaining edges are folded up. The thickness of the sheet is neglected.

1. Draw the net and express the height and the two dimensions of the base of the box. State the values of x which give a box of non-zero volume.

2. Express the volume $V(x)$ in cubic centimetres. Find the dimensions that make this volume maximum.

3. Check the values at both endpoints of the domain and give the exact maximum volume.



The four coloured areas are removed; the dotted lines show the folds. The drawing represents a illustrative value of x and does not give the optimum.

Exercise 6.6 · Stationary without extremum

A student claims: "If the derivative is zero at a point, the function must have an extremum local at this point." We study the function $f(x) = x^3$ to test this claim.

1. Calculate $f'(0)$.

2. Compare to $f(0)$ the values of $f(x)$ for real numbers x negative then positive, as close to 0 as we want.

3. Conclude about the existence of a local extremum at 0 and about the proposed statement.

Exercise 6.7 · Compare Strategies

We want to prove that, for every real number x, $x^2 - 6x + 12 \geq 3$.

1. Give a proof that makes a square appear.

2. Give a second proof by studying the variation of a function.

3. State the equality case. Compare the two methods: which one requires fewer calculations here?

Exercise 6.8 · Delayed review

The reciprocal function $f(x) = \dfrac{1}{x}$ has domain $D = \mathbb{R} \setminus \{0\}$. A student calculates $f\prime(x) = -\dfrac{1}{x^2} < 0$ and concludes that f is decreasing on the whole of D.

1. Identify the missing hypothesis in the application of the theorem linking derivative sign and variations.

2. State the two intervals on which the conclusion is correct.

3. Using the numbers -1 and 1, construct a counterexample to its conclusion on the whole D.

Independent work. Do only the task assigned at the end of the session, in the scheme of work Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not use up the whole reserve.

To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, Exercise 12 p. 156, Questions 1 and 2. Calculate the difference, study its sign and translate this into the relative position of the curves.

CQFD, exercise 16 p. 156, questions f and g. Give, for each function, an interval where it is both negative and increasing; justify both properties separately.

MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. $(x^3)'$ value
   - A. x^2
   - B. $3x^2$
   - C. $3x$
   - D. $\frac{x^4}{4}$
2. $(\frac{1}{x})'$ value
   - A. $-\frac{1}{x^2}$
   - B. $\frac{1}{x^2}$
   - C. $-\frac{1}{x}$
   - D. 1
3. $(uv)'$ value
   - A. $u'v'$
   - B. $u' + v'$
   - C. uv'
   - D. $u'v + uv'$
4. $f' > 0$ on an interval implies
   - A. f positive
   - B. f negative
   - C. f strictly increasing
   - D. f constant
5. $\frac{x^2+1}{x-1}$ is defined if
   - A. $x > 0$
   - B. $x \neq 1$
   - C. $x \neq -1$
   - D. $x \neq 0$

MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. $x^3 - x$ is

   A. even

   B. constant

   C. odd

   D. neither even nor odd

2. $f\prime(a) = 0$ necessarily implies

   A. horizontal tangent

   B. Maximum

   C. Minimum

   D. f constant

3. Derivative of $\sqrt{x}$:

   A. $2\sqrt{x}$

   B. $\dfrac{1}{\sqrt{x}}$

   C. $\dfrac{\sqrt{x}}{2}$

   D. $\dfrac{1}{2\sqrt{x}}$, $x > 0$

4. Minimum of $x + \dfrac{1}{x}$ on]0; + ∞[:

   A. 0

   B. 2

   C. 1

   D. none

5. Newton's method for $x^2 - 2$ is not defined with $x_0 =$

   A. 1

   B. 2

   C. 0

   D. -1

True or false? Justify.

1. The derivative of a positive function is positive.

2. An interior extremum of a differentiable function gives a zero derivative.

3. The product of two increasing functions is increasing.

Personal review

I differentiate on the correct domain; I simplify without losing excluded values; I distinguish extremum local and global; I sometimes choose a factorisation rather than a derivative.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.

Solutions and teaching notes, chapter 6

Teacher use. The following answers concern the tasks not solved in the lesson. Do first make the approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. 6. *Action:* Go back to rate 2 $a+h$.
2. $\mathbb{R}\setminus\{2\}$. *Action:* Set the denominator to zero.
3. Positive outside [-2; 1], negative between the roots, zero at the endpoints. *Action:* Construct two sign rows.
4. x^3-2x^2+x-2. *Action:* Distribute each term without leaving any out.
5. They are equal. *Action:* Substitute -x in brackets.
6. $y=2$. *Action:* Go back to the tangent equation and simplify.

Investigation

x^3 is strictly increasing with a horizontal tangent at 0. For a differentiable increasing function, the rates on both sides are positive or zero, so the derivative cannot be negative.

Direct applications

1. $5x^4$ on $\mathbb{R}$; $-\dfrac{2}{x^3}$ for $x\neq 0$; $\dfrac{1}{2\sqrt{x}}$ for $x>0$.
2. $2x(3x-2)+3(x^2+1)=9x^2-4x+3$; $-\dfrac{2x}{(x^2+1)^2}$ on $\mathbb{R}$.
3. $f\prime=2x-4$, so decreasing up to 2, then increasing. Minimum -3 at 2 ; values at the endpoints 1 and -2, so the maximum is 1 at 0.
4. First even, second odd; difference $x^2+1-2x=(x-1)^2\geq 0$, equality at 1.
5. $A(x)=x(6-x)$ on [0; 6] ; $A\prime=6-2x$. Maximum 9 m ^{2} at 3.

Exercises: methods and answers

Exercise 6.1 · A collection of derivatives. $12x^3-2$; $2x(x-1)+(x^2+2)=3x^2-2x+2$; $\dfrac{[2(x+3)-(2x+1)]}{(x+3)^2}=\dfrac{5}{(x+3)^2}$ for $x\neq -3$.

Exercise 6.2 · A complete table. $f\prime=3(x-1)(x-3)$; signs + , −, +. Values $f(0)=1$, $f(1)=5$, $f(3)=1$, $f(4)=5$. Minimum 1 at 0 and 3; maximum 5 at 1 and 4. The endpoints count.

Exercise 6.3 · A quotient. $f\prime=1-\dfrac{4}{x^2}=\dfrac{(x-2)(x+2)}{x^2}$. Negative sign before 2, positive after. Minimum 4 at 2; for $x>0$, $x+\dfrac{4}{x}\geq 4$, equality only at 2.

Exercise 6.4 · Compare two curves. $f-g=x(x-1)(x+1)$. Signs: negative on $]-\infty;-1[$, positive on $]-1;0[$, negative on $]0;1[$, positive on $]1;+\infty[$. Intersection at $-1, 0, 1$. We can do the table of the three factors without differentiation.

Exercise 6.5 · A package. $V=x(12-2x)^2=4x(6-x)^2$. $V'=12(x-2)(x-6)$, positive on $[0;2[$, negative on $]2;6[$. $V(0)=V(6)=0$, $V(2)$ = 128 cm ^{3}. Maximum at x=2 cm; box 8 $\times 8 \times 2$.

Exercise 6.6 · Stationary without extremum. $f'=3x^2$, so the value at 0 is 0. In every neighbourhood of 0, negative x values give $f(x)<0$ and positive x $f(x)>0$; no extremum.

Exercise 6.7 · Compare strategies. Square: $(x-3)^2+3\geq 3$. Derivative: $2x-6$, minimum at 3, value 3. The first avoids a complete study of variations; the second illustrates the link between chapters.

Exercise 6.8 · Delayed review. It decreases on $]-\infty;0[$ and $]0;+\infty[$. It does not decrease on their union: $-1<1$ but $f(-1)=-1<1=f(1)$. The domain is not an interval.

CQFD, Exercise 12 p. 156, Questions 1 and 2. $f(x)-g(x)=\frac{1}{2}x^2-1=\frac{1}{2}(x-\sqrt{2})(x+\sqrt{2})$. The curve of f lies strictly above that of g for $x<-\sqrt{2}$ or $x>\sqrt{2}$, below between these two numbers; equality at the x-values $\pm\sqrt{2}$.

CQFD, exercise 16 p. 156, questions f and g. For $f(x)=x^3$, we can choose $]-\infty;0[$: $f(x)<0$ and the function is strictly increasing. For $g(x)=x^2-4$, we can choose $[0;2[$: $g(x)<0$ and the function is strictly increasing on this interval. Subintervals not reduced to a single point are also suitable.

MCQ: answers and reasons

MCQ 1.

1. B. Integer power.

2. A. Rate of change of the reciprocal.

3. D. Product rule.

4. C. Sign of the derivative and variation.

5. B. Non-zero denominator.

MCQ 2.

1. C. $f(-x)=-f(x)$.

2. A. The other conclusions need more.

3. D. Formula and domain.

4. B. Attained at 1.

5. C. Derivative $2x_0=0$.

True or false

1. False: $f(x) = \dfrac{1}{x} > 0$ on $]0; +\infty[$, but $f\prime(x) < 0$.

2. True, with the stated assumptions; at the endpoints, the conclusion is not forced.

3. False without sign assumptions: on $[-2; -1]$, $u(x) = v(x) = x$ increase but x^2 decreases.

Differentiated teaching. If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language mastered, then require a precise mathematical sentence in French.